

Mark Farinas

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Technical Skills

Languages: Python, Kotlin, Dart, SQL, JavaScript, HTML/CSS, GDScript, C, VBA

Frameworks & Libraries: Jetpack Compose, Flutter, Ktor, JetBrains Exposed, Next.js, React, FastAPI, Retrofit

Databases & Infrastructure: PostgreSQL, PostGIS, SQLite, Supabase, Microsoft Access, NocoDB, Docker

Cloud & DevOps: Cloudflare (Workers, DNS, Email Routing), Vercel, Railway.io, Firebase, GitHub Actions, DNS/TLS administration

AI/ML: Ollama, Groq, RAG pipelines, vector embeddings, Whisper AI, LLM evaluation, agent-trajectory analysis, inter-rater agreement (Cohen's κ , Krippendorff's α)

Quality & Tools: pytest, mypy --strict, CI test matrices, Git, Figma

Experience

Cohere - Data Annotation Specialist, Software Engineering

Aug 2026 – Present

Independent Contractor

- Evaluate model-generated code and multi-step agent trajectories against structured rubrics - instruction adherence, correctness, edge-case handling, security, and completeness - producing written justifications used as training signal for large language model development
- Debug and review AI-generated code across Python, JavaScript, Java, Go, and SQL; assess tool selection, step ordering, and error recovery in agent execution traces

Alberta Federation of Rural Water Coops - Database Administrator Intern

May 2026 – Present

Edmonton, AB

- Architected a re-runnable Python ETL pipeline (pyodbc to psycpg) migrating a legacy Microsoft Access database to a containerized PostgreSQL + NocoDB stack, validated by a 125-test fidelity suite
- Designed and built TimeEntrySystem, a role-based Access/VBA timesheet application with a three-stage approval workflow and a three-layer audit/integrity model for anti-fraud assurance
- Administering the Water Federation website and automating standard workplace procedures, including form development and recurring administrative workflows
- Coordinating with stakeholders across rural Alberta water systems on data structure, access requirements, and migration rollout

Alberta Precision Labs - COVID-19 Surveillance Research

Dec 2022 – Dec 2023

Edmonton, AB

- Conducted COVID-19 surveillance research using qPCR and dPCR detection methods across multiple Albertan communities; executed daily lab workflows including RNA extraction, RT-PCR, and digital PCR

Popeye's Supplements - Customer Service

Jul 2022 – Present

Edmonton, AB

- Provide solution-based customer service, advising clients on supplements for athletic performance and wellness goals; conduct in-store demos and community education initiatives

Awards & Publications

1st Place - CMPT 395 Software Engineering Competition (IEEE-sponsored)

Apr 2026

Won both the Semi-Finals and the Grand-Finals with Scheduler

IEEE SmartEdu 2026 - co-authored full research paper

Accepted Jul 2026

“Unjamming Deferrals: Supporting Students Through JAMN Bot, a Mobile RAG-Grounded Policy Assistant”

Education

MacEwan University - BSc in Computer Science
Edmonton, AB

Jan 2024 – Jun 2028

University of Alberta - BSc in Immunology and Infection
Edmonton, AB

Sep 2016 – Apr 2021

Projects

Scheduler - Backend & Android Developer, 5-person Agile/Scrum team
Kotlin, Ktor, JetBrains Exposed, SQLite, Jetpack Compose, Docker, Ollama

Jan 2026 – Apr 2026

- Architected the Ktor + JetBrains Exposed + SQLite backend with JWT-based role authentication and a room-booking engine featuring conflict detection and proctor load-balancing, on a deployed multi-role exam-scheduling platform
- Contributed frontend screens in Kotlin + Jetpack Compose, including role-specific interfaces for students, faculty, and admin; integrated Retrofit REST client, Firebase Authentication, FCM push notification handling, and DataStore session management
- Designed automated FCM push notification triggers and a scheduled no-show poller in the backend service layer
- Built an Ollama-powered RAG pipeline for the AI policy assistant, combining nomic-embed-text vector embeddings with cosine-similarity retrieval over a SQLite vector store and a role-aware prompt builder
- Designed and ran the evaluation behind the resulting IEEE SmartEdu paper: built a multi-condition benchmark dataset, ran temperature ablation studies, and computed inter-rater agreement between human evaluators and an LLM-Judge
- Containerized with Docker and deployed on Railway.io with environment-driven secrets management; participated in bi-weekly sprints, code reviews, and cross-team conflict resolution

iMapped - Web Core & Release Engineering, 5-person venture team
Flutter, Dart, Supabase, PostGIS, Groq, Vercel, GitHub Actions

May 2026 – Present

- Built the web core of a trip-itinerary planner live at app.imapped.ca and shipped to both web and native Android from one codebase (88 Dart source files across 210 commits, 172 tests green)
- Implemented the client-side route optimizer in Dart - haversine distance with nearest-neighbour construction and 2-opt improvement - to sequence day-by-day plans and minimize total travel time
- Integrated the Flutter client with Supabase (Postgres + PostGIS, PKCE authentication, Edge Functions) behind repository interfaces, so seed-backed fakes could be swapped for live implementations one repository at a time without touching UI code
- Designed and implemented the recommendation re-ranker on Groq (Llama 70B) as a Supabase Edge Function, caching ranked results to the database to control both response latency and API cost
- Led UI/UX and user research across two rounds of user testing; findings drove a pivot from flight booking to itinerary planning, and a later repositioning from tourist-only to an everyday app with local loyalty mechanics
- Built as a 5-person venture team in MacEwan's technology-entrepreneurship course (COSL 301), owning the web core, landing page, and release engineering while teammates ran the backend, QA, and business/PM lanes
- Presented the product live twice - the course final (Jul 2026) and a follow-up stakeholder demo - walking non-technical audiences through the market positioning, the loyalty-partnership business model, and the deployed product itself
- Carried the venture past the course: the team elected to continue building after the term ended, with me owning the new GitHub organization, repository setup, and continuity of the live site during the transition

Retail Storefront Rescue & Migration - Solo client engagement
Cloudflare Workers, DNS/TLS, GoDaddy

Jul 2026

- Rebuilt and relaunched a local retailer's storefront site overnight, reconnecting customers to the business's products and online store after a multi-day outage had severed its primary customer-facing channel - delivered at \$0/month recurring cost
- Kept the client's revenue-generating online store up with zero downtime through the entire nameserver cutover: enumerated the legacy cPanel zone, verified DNSSEC state beforehand to avoid a total resolution failure, and preserved the third-party e-commerce subdomain untouched

- Rebuilt the site from scratch after determining the only surviving artifact was an unusable dev-mode SPA capture (dev-server instrumentation, JavaScript-dependent hidden content, third-party session recording, and a `noindex` directive); delivered a dependency-free static site with no build step
- Self-directed the migration across platforms I had not used before (GoDaddy domain administration, Cloudflare Workers, DNS and Email Routing), researching provider terms and free-tier commercial-use restrictions to confirm the platform was a lawful, sustainable fit before committing the client to it
- Configured TLS, Cloudflare Email Routing, and SPF/DKIM/DMARC (`p=reject`) so customer enquiries reach the business and its domain cannot be spoofed; added a fails-closed 18+ age gate and the privacy and terms pages the previous site had linked to but never implemented
- Structured the handover around client ownership - client-owned domain and administrator access, a portable static-file deliverable, and a written runbook - so the business can change hosts or developers without ever losing its storefront again

VocabMaxxing [🔗](#) - Android, Backend & UX Lead

Feb 2026 – Jul 2026

Kotlin, Jetpack Compose, Ktor, Groq, Firebase, Figma

- Implemented the Android frontend (login, dashboard, daily practice with microphone capture) in Kotlin + Jetpack Compose
- Built the backend on Ktor + JetBrains Exposed + SQLite with Firebase Admin token verification and REST endpoints for auth, attempts, and user dashboards; containerized with Docker and deployed on Railway.io
- Designed a Groq-powered (Llama 3.3 70B) semantic scoring service with a multi-dimensional grading rubric and hardened prompt-injection defenses
- Developed a custom algorithmic scoring engine combining morphological stemming and vocabulary-diversity heuristics, combined with the AI scorer for a hybrid evaluation pipeline
- Led ideation and UX design; produced low- and mid-fidelity Figma wireframes across multiple iterations covering layout, colour scheme, navigation flow, and gamification mechanics (streaks, XP, coin shop)

Phone-Menu-Transcriber [🔗](#) - Solo

Jun 2026

Python, Whisper AI, Ollama, FastAPI, Pydantic

- Built a packaged Python transcription service pairing Whisper speech-to-text with a locally-hosted Qwen 3 8B (Ollama) extractor behind a pluggable interface
- Produced schema-constrained JSON output via Pydantic, exposed as both a CLI and a FastAPI service, with `mypy --strict` typing and a four-version CI matrix

Maintenance Calorie Finder [🔗](#) - Solo

Python, SciPy, pytest

- Built a Python CLI application that reverse-engineers maintenance calories from tracked intake and weight data; models metabolic adaptation over a goal window and solves for target intake with SciPy bounded optimization, validated by a 24-test pytest suite

Personal Portfolio [🔗](#) - Solo

Next.js, React

- Developed a personal portfolio website using Next.js with a design system inspired by American Psycho's business card aesthetic, featuring client-side routing, responsive navigation, and interactive hover states

TiMED - Hack4Health 2025, co-designer

2025

Hardware + software concept, Figma

- Ideated and co-designed a hardware-software medication management system; designed companion app wireframes featuring daily medication schedules, adherence reports, and QR-based device pairing; presented the proof-of-concept to a panel of judges